Decision Tree (DT) classifier using the Iris dataset

Python codes for Decision Tree (DT) classifier using the Iris dataset and then testing it with an unknown sample (Jupyter Notebook)

1. Import Libraries

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report
```

2. Load the Iris Dataset

In [2]:
```
iris = load_iris()

X = iris.data
y = iris.target

print("Feature names:", iris.feature_names)
print("Target names:", iris.target_names)
```

```
Feature names: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal
width (cm)']
Target names: ['setosa' 'versicolor' 'virginica']
```

3. Split the Dataset

In [3]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Training samples:", X_train.shape[0])
print("Testing samples:", X_test.shape[0])
```

```
Training samples: 120
Testing samples: 30
```

4. Train the DT Classifier

In [4]:
```
dt = DecisionTreeClassifier(criterion='gini', random_state=42)
dt.fit(X_train, y_train)
```

Out[4]:
```
DecisionTreeClassifier
DecisionTreeClassifier(random_state=42)
```

5. Evaluate the Model

In [5]:
```
y_pred = dt.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n",
      classification_report(y_test, y_pred, target_names=iris.target_names))
```

```
Accuracy: 1.0

Classification Report:
               precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       1.00      1.00      1.00         9
   virginica       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```

6. Predict with Unknown Data

In [6]:
```
# Example unknown data (sepal length, sepal width, petal length, petal width)
unknown_sample = np.array([[5.1, 3.5, 1.4, 0.2]])

prediction = dt.predict(unknown_sample)
predicted_class = iris.target_names[prediction[0]]

print("Predicted class for unknown sample:", predicted_class)
```

```
Predicted class for unknown sample: setosa
```

7. Try Multiple Unknown Samples

In [7]:
```
unknown_samples = np.array([
    [6.7, 3.1, 4.7, 1.5],  # likely versicolor
    [7.2, 3.6, 6.1, 2.5],  # likely virginica
    [5.0, 3.4, 1.5, 0.2]   # likely setosa
])

predictions = dt.predict(unknown_samples)

for i, pred in enumerate(predictions):
    print(f"Sample {i+1} → {iris.target_names[pred]}")
```

```
Sample 1 → versicolor
Sample 2 → virginica
Sample 3 → setosa
```

8. Visualize the DT

In [8]:
```
from sklearn.tree import plot_tree

plt.figure(figsize=(12, 8))
plot_tree(
    dt,
    feature_names=iris.feature_names,
    class_names=iris.target_names,
    filled=True
)
plt.title("Decision Tree for Iris Dataset")
plt.show()
```

